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Explainable and Verified Capacity Recovery for a Two-Node Cloud Hypervisor VM Controller

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Problem Description

Abstract

CleanSlot SafeCore is a lightweight controller that manages VMs across two hosts using Cloud Hypervisor. It combines memory-aware placement with explainable failures, safe cleanup and verified recovery.

issues.

Problem Description

- Generic errors make resource shortages difficult to diagnose.
- Failed VM operations may leave orphaned processes.
- Unsafe cleanup can affect unrelated workloads.
- Controller restarts may leave stale VM records.

Goal: Enable intelligent placement and safe, verifiable VM recovery.

Background

- KVM: Enables hardware-assisted virtualisation on Linux.
- Cloud Hypervisor: Creates and manages virtual machines.
- Controller: Selects hosts and coordinates VM operations.
- Worker agents: Execute operations on each host.
- State reconciliation: Detects differences between recorded and actual VM states.

Motivation: Reliable VM management requires resource awareness, accurate state tracking and safe failure recovery.

Literature & RFC Review

Literature & RFC Review

- Cloud Hypervisor API: VM lifecycle management.
- Proxmox VE: Existing virtualisation management.
- OpenStack Nova: Resource-aware VM scheduling.
- Kubernetes: Scheduling and state reconciliation concepts.
- RFC 6750: Bearer-token authentication.
- RFC 9457: Standardised API error responses.

Identified opportunity: Combine explainable capacity failures, authorised cleanup and verified recovery in a lightweight two-host controller.

[1] Cloud Hypervisor API Documentation
[2] RFC 9457 — Problem Details for HTTP APIs

Proposed Solution

Proposed Solution — CleanSlot SafeCore
PLACE — Select a suitable host using resource availability and admission checks.
PROTECT — Prevent automatic termination of running VMs.
EXPLAIN — Show why a VM request failed through the Capacity Rescue View.
RECOVER — Enable authorised, targeted cleanup.
PROVE — Verify cleanup and generate a recovery receipt.
Core features: Two workers, authenticated APIs, VM lifecycle management, resource limits and restart reconciliation.

Optimization Proposed

CleanSlot SafeCore improves traditional VM controllers by providing explainable capacity errors, resource-aware host selection, verified targeted cleanup, restart reconciliation, and auditable recovery receipts. Its effectiveness will be evaluated using placement accuracy, cleanup correctness, recovery time, operation latency, and state consistency, with results measured through testing.

Solution Architecture & Design

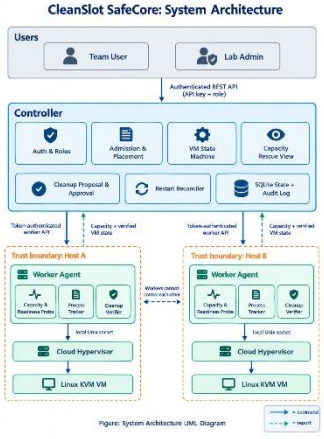


Figure: System Architecture UML Diagram

Data Flow Diagram

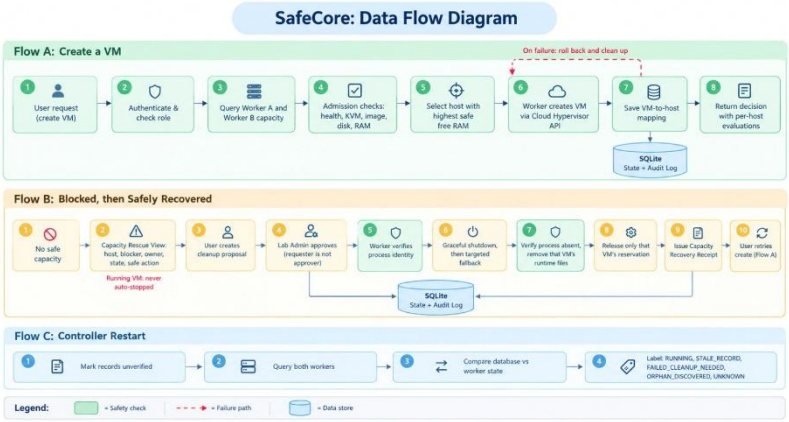


Figure: Data Flow Diagram (DFD)

Deployment Timeline

1. Environment: Configure two Linux/KVM hosts and Cloud Hypervisor.
2. Core Development: Build workers, controller and VM lifecycle APIs.
3. Scheduling: Implement memory-aware placement and resource limits.
4. Safe Recovery: Add cleanup verification, approval and audit logging.
5. Testing: Validate failures, restart recovery and two-host operations.
6. Submission: Finalise report, demonstration and GitHub repository.

Methodology: Incremental development with testing at every phase.

Bibliography I

[1] Cloud Hypervisor API Documentation
https://github.com/cloud-hypervisor/cloud-hypervisor/blob/main/docs/api.md
[2] Proxmox VE Documentation
https://pve.proxmox.com/pve-docs/
[3] OpenStack Nova Documentation
https://docs.openstack.org/nova/latest/
[4] IETF RFC 6750 — Bearer Token Usage
https://www.rfc-editor.org/rfc/rfc6750
[5] IETF RFC 9457 — Problem Details for HTTP APIs
https://www.rfc-editor.org/rfc/rfc9457

Thank You for Your Attention!